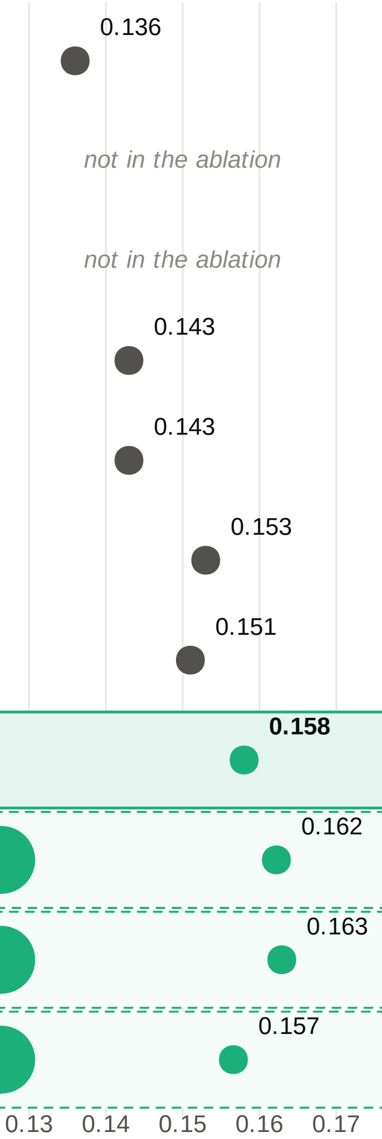
The versions of CLR we tested, from raw dependence to the full method and its exploratory extensions

E. coli: strong-evidence regulons, 466 experiments (RegulonDB). K562: CORUM complexes. Single cells: BEELINE, median over 14 dataset-and-gene-set cases.

Version	Dependence	Context normalization	Softmax	Walk	Log + second CLR	E. coli AUPR (SC)	K562 CORUM (× random)	BEELINE median ratio
CLR context normalization, scores ranked directly	mutual information	CLR z	—	—	—	0.136	6.39	1.48
CLR-NS CLR on |r| of normal scores (Blom)	|r| of normal scores	CLR z	—	—	—	*not in the ablation*		**1.60**
|r| (correlation) no normalization, scores ranked directly	|r|	—	—	—	—	*not in the ablation*	7.00	1.20
Attention on |r| softmax + walk, no CLR normalization	|r|	—	softmax	t* = 8	—	0.143		
Attention on mutual information softmax + walk, no CLR normalization	mutual information	—	softmax	t* = 6	—	0.143		
Attention on CLR of |r| CLR normalization, then softmax + walk	|r|	CLR z	softmax	t* = 7	—	0.153		
Attention on CLR-NS CLR-NS scores, then softmax + walk	|r| of normal scores	CLR z	softmax	t* = 5	—	0.151		no gain over CLR-NS
CLR attention (adopted) MI, CLR normalization, softmax + walk	mutual information	CLR z	softmax	t* = 9	—	**0.158**	**3.51**	
a3r (exploratory) attention, then log of the walk + a second CLR	mutual information	CLR z	softmax	t = 9	log + CLR	0.162	**7.95**	
a3r2: two layers (exploratory) a3r fed back as the score of a second layer	mutual information	CLR z	softmax	t = 9 ×2	log + CLR	0.163	**8.19**	
a3r3: three layers (exploratory) a3r fed back twice	mutual information	CLR z	softmax	t = 9 ×3	log + CLR	0.157	**8.15**	



t* = held-out depth, chosen per version; the method comparison itself fixes t = 9. Exploratory rows: t = 9 in every layer, not re-selected; none was adopted (26 Oct preregistered test, 12 Nov stacked-layers run).

a3r was picked among about 20 read-outs by its K562 scores (a selection, not a fit: it has no fitted parameters). Gaps between a3r, a3r2 and a3r3 are inside the noise. BEELINE: not run for a3r.

Compared with (BEELINE median ratio): GENIE3 1.69 PIDC 1.56 GRNBoost2 1.47 scPRINT zero-shot attention: CLR-NS ahead in 7 of 7 datasets (+1.03 log2 AUPR)